

Lecture-14

Mathematics 1 (15B11MA111)

CO [C105.2]

Module: Double Integrals

Topic: Change of order

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Bali, Goyal ,“A Textbook of Engineering Mathematics”, Laxmi Publications.

CHANGE OF ORDER OF INTEGRATION

If the limits of the integration are variables, a change in the order of integration necessitates change in the limits of integration. A rough sketch of the region of integration helps in fixing the new limits of integration.

Working rule. When it is required to change the order of integration in an integral for which limits are given, we first find out the region of integration from the limits. When region R of integration is known, we next assign limits of integration in the reverse order from geometric considerations.

Example 1. Evaluate by changing the order of integration

$$\int_0^1 \int_{e^x}^e \frac{dydx}{\log y} .$$

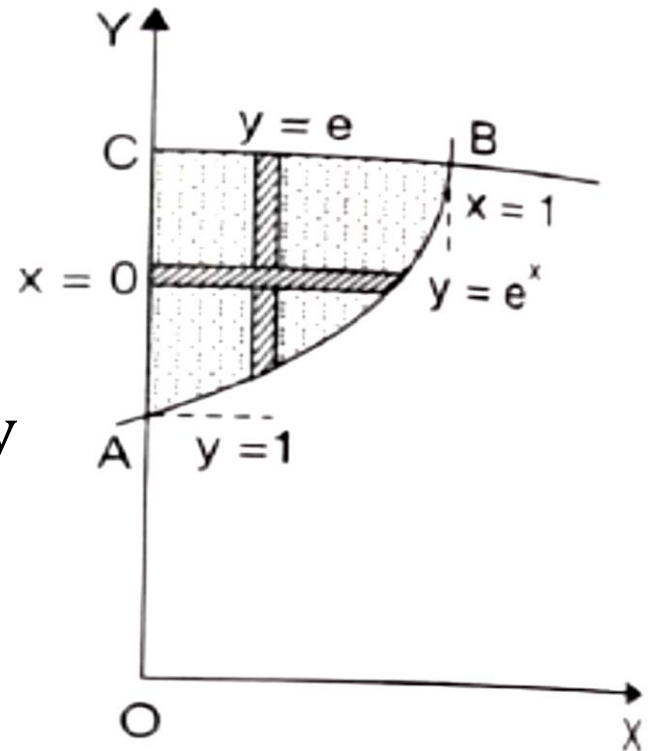
Sol. The given limits show that the region of integration is bounded by the curves

$$y = e^x, y = e, x = 0, x = 1$$

It is the area shaded in the diagram.

It can also be considered as bounded by

$$x = 0, x = \log y, y = 1, y = e$$



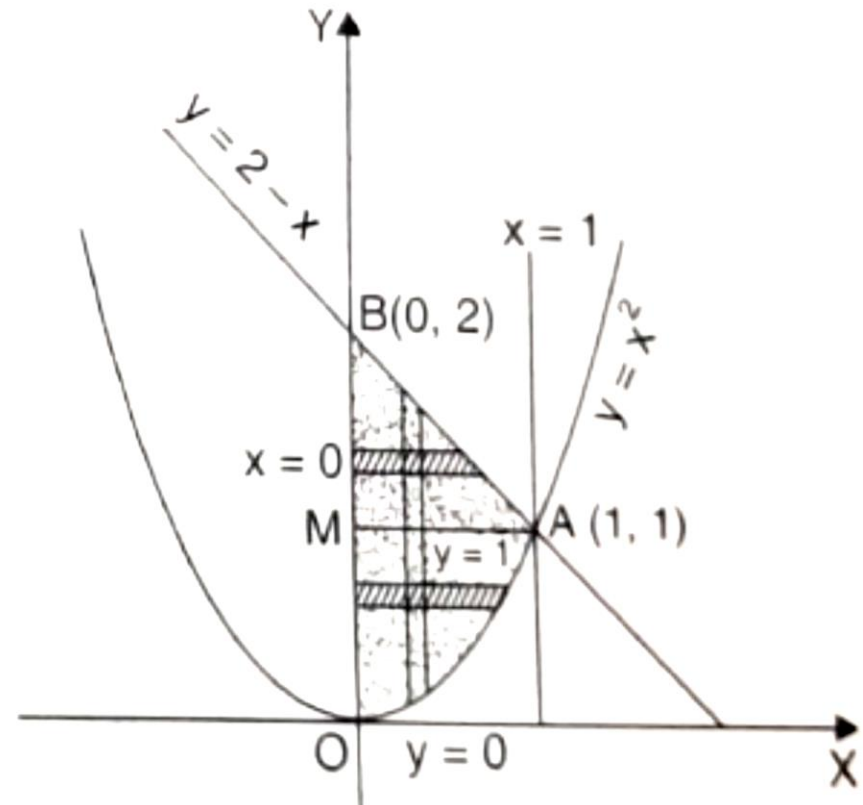
Hence,

$$\begin{aligned}\int_0^1 \int_{e^x}^e \frac{dydx}{\log y} &= \int_1^e \int_0^{\log y} \frac{1}{\log y} dx dy \\&= \int_1^e \left(\frac{x}{\log y} \right)_0^{\log y} dy \\&= \int_1^e 1 \cdot dy = (y)_1^e = e - 1.\end{aligned}$$

Example 2. Change the order of integration in $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$ and hence evaluate.

Sol. From the limits of integration, it is clear that we have to integrate first with respect to y which varies from $y = x^2$ to $y = 2 - x$ and then with respect to x which varies from $x = 0$ to $x = 1$.

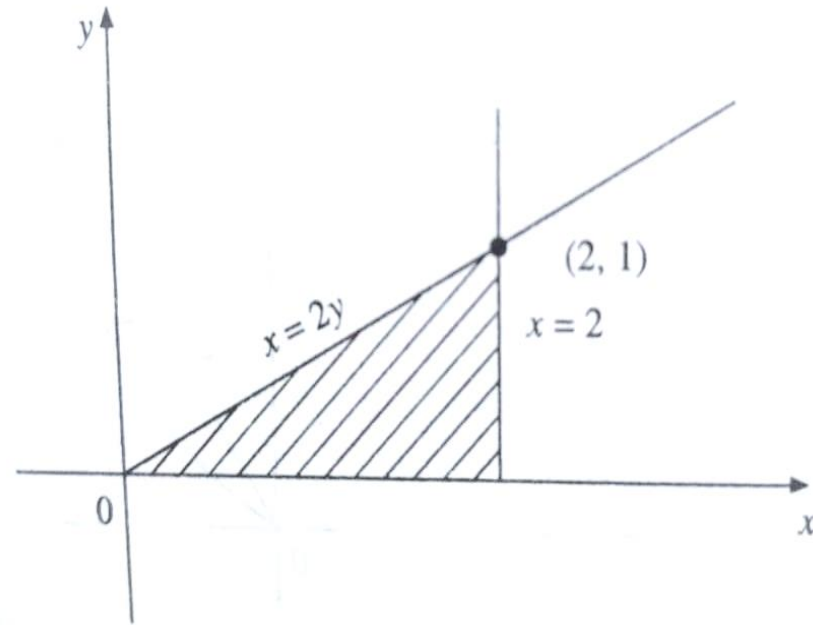
For changing the order of integration, we divide the region of integration into horizontal strips.



$$\begin{aligned}
I &= \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy + \int_1^2 \int_0^{2-y} xy \, dx \, dy \\
&= \int_0^1 y \cdot \left[\frac{x^2}{2} \right]_0^{\sqrt{y}} dy + \int_1^2 y \cdot \left[\frac{x^2}{2} \right]_0^{2-y} dy \\
&= \frac{1}{2} \int_0^1 y^2 \, dy + \frac{1}{2} \int_1^2 y(2-y)^2 \, dy \\
&= \frac{1}{2} \left[\frac{y^3}{3} \right]_0^1 + \frac{1}{2} \left[2y^2 - \frac{4y^3}{3} + \frac{y^4}{4} \right]_1^2 \\
&= \frac{1}{6} + \frac{1}{2} \left(\frac{5}{12} \right) = \frac{3}{8}.
\end{aligned}$$

Example 3. evaluate the double integral $\iint_R e^{x^2} dx dy$, where the region R is given by $R: 2y \leq x \leq 2$ and $0 \leq y \leq 1$.

Sol. The integral cannot be evaluated by integrating first with respect to x . We try to evaluate it by integrating it first with respect to y . The region of integration is given in figure. We have



$$I = \int_0^2 \left[\int_0^{x/2} e^{x^2} dy \right] dx = \frac{1}{2} \int_0^2 x e^{x^2} dx = \frac{1}{4} (e^4 - 1)$$

Application of double integration to find area

$\iint_R dx \, dy$ gives the area A of the region R .

For example, if R is the rectangle bounded by the lines

$x = a, x = b, y = c$ and $y = d$, then

$$A = \int_c^d \int_a^b dx \, dy = \int_c^d \left[\int_a^b dx \right] dy = (b - a) \int_c^d dy = (b - a)(d - c)$$

gives the area of the rectangle.

Example 4. Find the area of the region bounded by the curves $y = x^2$ and $y = x + 2$.

Sol. The area of the region is given by $A = \iint_R dx dy$

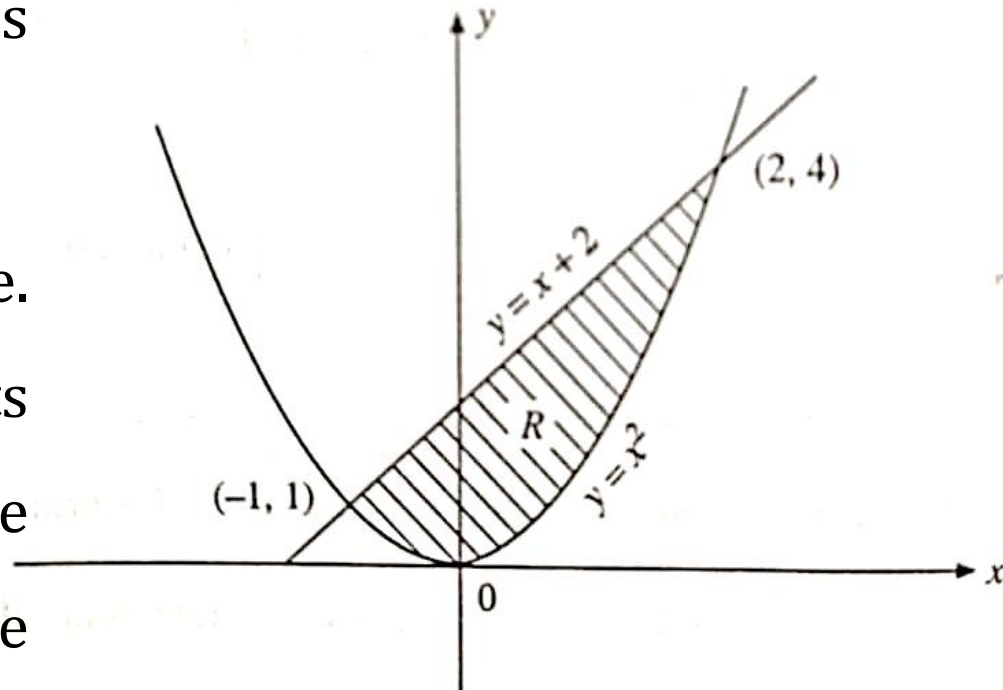
The region is given in Figure.

The line $y = x + 2$ intersects the parabola $y = x^2$ at the points $(-1, 1)$ and $(2, 4)$. The

limits of the integration can be

written as $-1 \leq x \leq 2, x^2 \leq$

$y \leq x + 2$.



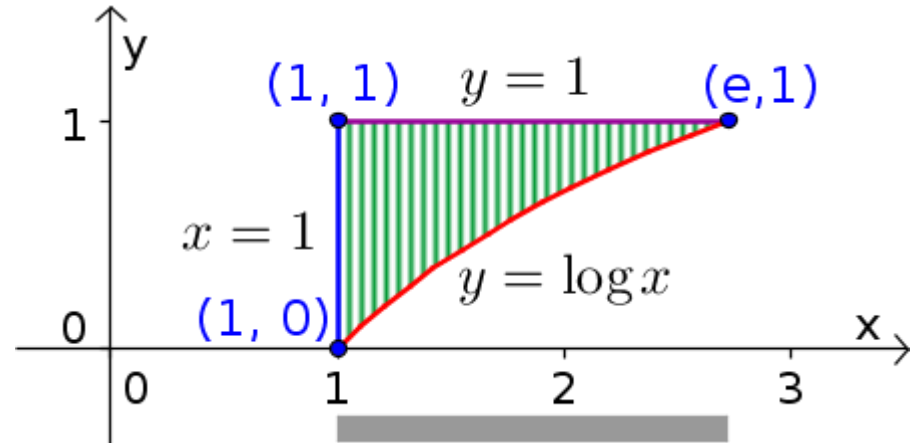
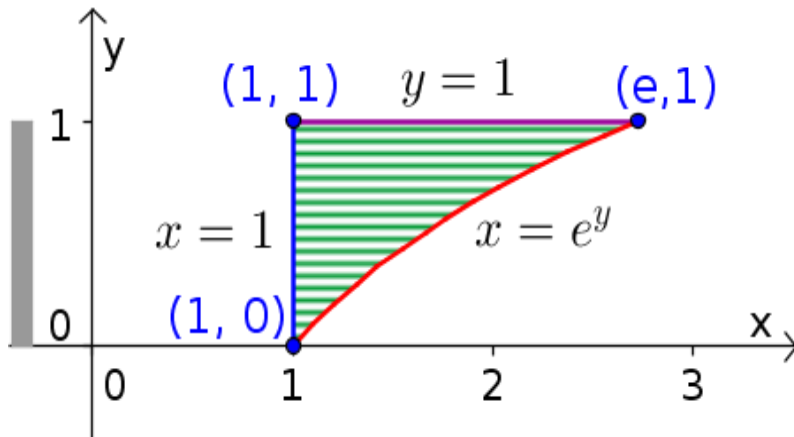
$$A = \int_{-1}^2 \left[\int_{x^2}^{x+2} dy \right] dx = \int_{-1}^2 (x + 2 - x^2) dx$$

$$= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^2$$

$$= \left(-\frac{9}{3} + \frac{3}{2} + 6 \right)$$

$$= \frac{9}{2}.$$

1. Change the order of integration $\int_0^1 \int_1^{e^y} f(x, y) dx dy$.



$$\int_1^e \left\{ \int_{\log x}^1 f(x, y) dy \right\} dx$$

THANK YOU